

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Chromium(III) 2-ethylhexanoate, 65-70% in mineral spirits

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>S33703</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

|                                                      |             |
|------------------------------------------------------|-------------|
| Skin Corrosion/Irritation                            | Category 2  |
| Serious Eye Damage/Eye Irritation                    | Category 1  |
| Carcinogenicity                                      | Category 2  |
| Reproductive Toxicity                                | Category 1B |
| Specific target organ toxicity - (repeated exposure) | Category 2  |

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



Signal Word

Danger

**Hazard Statements**

H351 - Suspected of causing cancer

H360 - May damage fertility or the unborn child

H315 - Causes skin irritation

H318 - Causes serious eye damage

H373 - May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements**

**Prevention**

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P308 + P313 - IF exposed or concerned: Get medical advice/attention

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

This product does not contain any known or suspected endocrine disruptors

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component                                  | CAS No     | Weight % |
|--------------------------------------------|------------|----------|
| Hexanoic acid, 2-ethyl-, chromium(3+) salt | 3444-17-5  | 68       |
| Fuels, diesel, no. 2                       | 68476-34-6 | 28       |
| 2-Ethylhexanoic acid                       | 149-57-5   | 4        |

## Section 4 - First Aid Measures

**Description of first aid measures**

**General Advice**

If symptoms persist, call a physician.

**New Zealand Emergency Tel.**

CHEMTREC®  
09 980 6780 or +64 9 980 6780

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

|                                     |                                                                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Ingestion                           | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| Self-Protection of the First Aider  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |
| First Aid Facilities                | Eyewash, safety shower and washroom.                                                                                                             |
| Most important symptoms and effects | None reasonably foreseeable. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                 |
| Notes to Physician                  | Treat symptomatically. Symptoms may be delayed.                                                                                                  |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

### Hazardous Combustion Products

Carbon oxides, Chromium oxide.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

#### Environmental Precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system.

#### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

#### Precautions to prevent secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### Advice on safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do

not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities**

**Storage Conditions**

Keep container tightly closed in a dry and well-ventilated place.

**Incompatible Materials**

None known.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Control parameters**

**Exposure limits**

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component                                  | New Zealand WEL | Australia                  | ACGIH TLV                          | The United Kingdom                                                    |
|--------------------------------------------|-----------------|----------------------------|------------------------------------|-----------------------------------------------------------------------|
| Hexanoic acid, 2-ethyl-, chromium(3+) salt |                 | TWA: 0.5 mg/m <sup>3</sup> |                                    | STEL: 1.5 mg/m <sup>3</sup> 15 min<br>TWA: 0.5 mg/m <sup>3</sup> 8 hr |
| Fuels, diesel, no. 2                       |                 |                            | TWA: 100 mg/m <sup>3</sup><br>Skin |                                                                       |
| 2-Ethylhexanoic acid                       |                 |                            | TWA: 5 mg/m <sup>3</sup>           |                                                                       |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls**

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

**Eye Protection**

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material             | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Viton (R). | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                  |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains.                                                                                                                                                                                                                                                                                       |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Green                    |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | Not applicable           |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | 265 °C / 509 °F          |                                          |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | 110 °C / 230 °F          | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | No information available |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| 2-Ethylhexanoic acid                           | 2.7                      |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 1.01                     |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### Other information

## Section 10 - Stability and Reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Sensitivity to Mechanical Impact</b> | No information available                   |
| <b>Sensitivity to Static Discharge</b>  | No information available                   |

|                                         |                                |
|-----------------------------------------|--------------------------------|
| <b>Hazardous Polymerization</b>         | No information available.      |
| <b>Hazardous Reactions</b>              | None under normal processing.  |
| <b>Conditions to Avoid</b>              | Heat, flames and sparks.       |
| <b>Incompatible Materials</b>           | None known.                    |
| <b>Hazardous Decomposition Products</b> | Carbon oxides. Chromium oxide. |

## **Section 11 - Toxicological Information**

### **Acute Effects**

### **Information on likely routes of exposure**

#### **Product Information**

|                   |                                    |
|-------------------|------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure. |
| <b>Eyes</b>       | Avoid contact with eyes.           |
| <b>Skin</b>       | Avoid contact with skin.           |
| <b>Ingestion</b>  | May be harmful if swallowed.       |

### **Numerical measures of toxicity**

#### **(a) acute toxicity;**

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Based on available data, the classification criteria are not met |
| <b>Dermal</b>     | Based on available data, the classification criteria are not met |
| <b>Inhalation</b> | Based on available data, the classification criteria are not met |

### **Toxicology data for the components**

| <b>Component</b>                           | <b>LD50 Oral</b>          | <b>LD50 Dermal</b>           | <b>LC50 Inhalation</b>                                     |
|--------------------------------------------|---------------------------|------------------------------|------------------------------------------------------------|
| Hexanoic acid, 2-ethyl-, chromium(3+) salt |                           | LD50 > 2000 mg/kg ( Rat )    |                                                            |
| Fuels, diesel, no. 2                       |                           | LD50 > 2000 mg/kg ( Rabbit ) | LC50 = 3.6 mg/L ( Rat ) 4 h<br>LC50 = 5.4 mg/L ( Rat ) 4 h |
| 2-Ethylhexanoic acid                       | LD50 = 1600 mg/kg ( Rat ) | LD50 = 1140 mg/kg ( Rabbit ) |                                                            |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

#### **(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available

#### **(f) carcinogenicity;**

Category 2  
There are no known carcinogenic chemicals in this product

#### **(g) reproductive toxicity;**

Category 1B

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity**

Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component            | Freshwater Fish                                               | Water Flea                                | Freshwater Algae                                                                                             | Microtox                                          |
|----------------------|---------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Fuels, diesel, no. 2 | LC50: = 35 mg/L, 96h<br>flow-through<br>(Pimephales promelas) |                                           |                                                                                                              |                                                   |
| 2-Ethylhexanoic acid | LC50: = 70 mg/L, 96h<br>(Pimephales promelas)                 | EC50: = 85.4 mg/L, 48h<br>(Daphnia magna) | EC50: = 41 mg/L, 96h<br>(Desmodesmus<br>subspicatus)<br>EC50: = 61 mg/L, 72h<br>(Desmodesmus<br>subspicatus) | EC50 = 110 mg/L 17 h<br>EC50 = 670 mg/L 30<br>min |

**Terrestrial ecotoxicity** There is no data for this product

**Persistence and Degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary

**Persistence** May persist.

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Product has a high potential to bioconcentrate

| Component            | log Pow | Bioconcentration factor (BCF) |
|----------------------|---------|-------------------------------|
| 2-Ethylhexanoic acid | 2.7     | No data available             |

**Mobility** No information available.

### Other adverse effects

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

#### **Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## Section 14 - Transport Information

### NZS 5433:2020

Not regulated

### IATA

Not regulated

### IMDG/IMO

Not regulated

#### **Environmental hazards**

No hazards identified

#### **Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable, packaged goods

#### **Special Precautions**

No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

#### **Additional information**

None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### **National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### **Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

### International Regulations



**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions  
according to EU REACH**

| Component            | REACH (1907/2006) - Annex XIV -<br>Substances Subject to<br>Authorization | REACH (1907/2006) - Annex XVII -<br>Restrictions on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 - Candidate<br>List of Substances of Very High<br>Concern (SVHC) |
|----------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Fuels, diesel, no. 2 | -                                                                         | Use restricted. See item 75.<br>(see link for restriction details)                  | -                                                                                                              |
| 2-Ethylhexanoic acid | -                                                                         | Use restricted. See item 75.<br>(see link for restriction details)                  | -                                                                                                              |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                     | CAS No     | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL           | IECSC | TCSI |
|-----------------------------------------------|------------|-------|------|-----------|--------|-----|----------------|-------|------|
| Hexanoic acid, 2-ethyl-,<br>chromium(3+) salt | 3444-17-5  | X     | X    | 222-357-3 | -      | -   | KE-14-006<br>0 | X     | X    |
| Fuels, diesel, no. 2                          | 68476-34-6 | X     | X    | 270-676-1 | -      | -   | KE-17287       | X     | X    |
| 2-Ethylhexanoic acid                          | 149-57-5   | X     | X    | 205-743-6 | -      | -   | KE-13740       | X     | X    |

| Component                                     | CAS No     | TSCA | TSCA Inventory<br>notification -<br>Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-----------------------------------------------|------------|------|-----------------------------------------------------|-----|------|-------|------|------|
| Hexanoic acid, 2-ethyl-,<br>chromium(3+) salt | 3444-17-5  | X    | ACTIVE                                              | X   | -    | X     | -    | -    |
| Fuels, diesel, no. 2                          | 68476-34-6 | X    | ACTIVE                                              | X   | -    | X     | -    | -    |
| 2-Ethylhexanoic acid                          | 149-57-5   | X    | ACTIVE                                              | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

**Legend**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

|                                                                                     |                                                                                    |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail |
| <b>LD50</b> - Lethal Dose 50%                                                       | <b>LC50</b> - Lethal Concentration 50%                                             |
| <b>EC50</b> - Effective Concentration 50%                                           | <b>ATE</b> - Acute Toxicity Estimate                                               |
| <b>WEL</b> - Workplace Exposure Limit                                               | <b>RPE</b> - Respiratory Protective Equipment                                      |
| <b>DNEL</b> - Derived No Effect Level                                               | <b>NOEC</b> - No Observed Effect Concentration                                     |
| <b>POW</b> - Partition coefficient Octanol:Water                                    | <b>BCF</b> - Bioconcentration factor                                               |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                 | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                    |
| <b>VOC</b> - (Volatile Organic Compound)                                            |                                                                                    |

**Key literature references and sources for data**

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS  
EPA Guide to classifying hazardous substances in New Zealand  
EPA - Assigning a product to an existing HSNO approval guide

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.  
Chemical incident response training.

|                         |                 |
|-------------------------|-----------------|
| <b>Revision Date</b>    | 08-Sep-2023     |
| <b>Revision Summary</b> | Initial Release |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**